Investigation of MZIs fabricated on a silicon photonics platform

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***Abstract*—This report presents the design of Mach-Zehnder interferometers on silicon-on-insulator platform. Waveguides in the strip configuration are used. Simulated results and measurement results are presented to understand the circuit performance.**

***Index Terms*—MZI, arm difference, waveguide, Si photonics**

I. Introduction

THE Mach–Zehnder interferometer is a device used to determine the relative phase shift between two coherent beams derived by splitting light from a single source. It has been widely used in many areas including optical communication, biosensing and chemical sensing, and quantum optics. Understanding the performance and loss of MZI devices is crucial for designing versatile applications. Here in this article, we investigate the performance of MZIs built using a directional coupler and a Y-branch coupler.

II. Theoretical model of MZI

For a Y-branch splitter, considering input light with intensity I_i and electric field E_i, the outputs are

$$I_1 = \frac{I_i}{2}, \qquad I_2 = \frac{I_i}{2}, \tag{1}$$

and

$$E_1 = \frac{E_i}{\sqrt{2}}, \qquad E_2 = \frac{E_i}{\sqrt{2}}. \tag{2}$$

For a Y-branch combiner, considering input fields E_1 and E_2, the output field is

$$E_o = \frac{E_1 + E_2}{\sqrt{2}}, \tag{3}$$

and the output intensity is

$$I_o = I_1 + I_2. \tag{4}$$

For an MZI configured with a Y-branch splitter and a Y-branch combiner, let the two arm lengths be L_1 and L_2. The propagation constant is β, and the loss is α. Then the output electric field is

$$E_o = \frac{E_i}{2}\left(e^{-i\beta L_1} + e^{-i\beta L_2}\right). \tag{5}$$

The propagation constant including loss is written as

$$\beta = \frac{2\pi n_{\text{eff}}}{\lambda} + i\frac{\alpha}{2}. \tag{6}$$

The output intensity is

$$I_o = \frac{E_i^2}{4}\left|e^{-i\beta L_1} + e^{-i\beta L_2}\right|^2. \tag{7}$$

Thus, the transfer function of the MZI is

$$T = \frac{1}{4}\left|e^{-i\beta L_1} + e^{-i\beta L_2}\right|^2 = \frac{1}{2}\left[1 + \cos(\beta\Delta L)\right], \tag{8}$$

where

$$\Delta L = L_1 - L_2. \tag{9}$$

For an MZI configured with one Y-branch splitter and one directional coupler, the two inputs to the directional coupler, which are the two outputs of the Y-branch splitter, are

$$E_1 = \frac{E_i}{\sqrt{2}}, \qquad E_2 = \frac{E_i}{\sqrt{2}}. \tag{10}$$

Considering an ideal 2×2 splitter made with a broadband directional coupler and two arm lengths L_1 and L_2, the output fields are

$$E_{o1} = \frac{E_i}{2}\left(e^{j\beta L_1} + je^{j\beta L_2}\right), \tag{11}$$

$$E_{o2} = \frac{E_i}{2}\left(je^{j\beta L_1} + e^{j\beta L_2}\right). \tag{12}$$

The corresponding output intensities are

$$I_{o1} = \frac{I_i}{2}\left[1 - \sin\big(\beta(L_2 - L_1)\big)\right], \tag{13}$$

$$I_{o2} = \frac{I_i}{2}\left[1 + \sin\big(\beta(L_2 - L_1)\big)\right]. \tag{14}$$

Therefore, the transfer function is

$$T = \frac{1}{2}\left(1 + \sin(\beta\Delta L)\right). \tag{15}$$

III. Simulation

A. Waveguide

In our design, a strip waveguide with a width of $500\,\text{nm}$ and a height of $220\,\text{nm}$ is used. The simulated fundamental TE and TM mode profiles are shown in Fig. 1. It can be seen that the TM mode extends further into the cladding layer. This is because the vertical confinement of the TM mode is weaker in a $220\,\text{nm}$-thick silicon waveguide. In particular, the dominant electric-field component of the TM mode is normal to the core–cladding interface, and the continuity condition of the electric displacement field leads to a relatively larger field penetration into the low-index cladding. As a result, the TM mode has a lower confinement in the silicon core and a larger evanescent tail in the cladding compared with the TE mode.

The effective index and group index of the TE mode as functions of wavelength are shown in Fig. 2. At $1550\ \text{nm}$, $n_g = 4.2037$ for TE mode. From the figure, it can be observed that the effective index decreases with increasing wavelength,

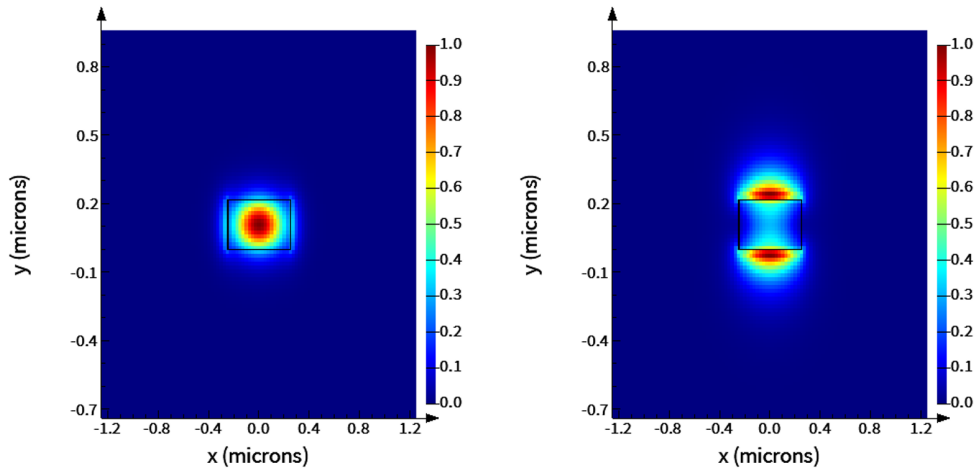

Fig. 1. Electric field intensity of TM mode in the waveguide.

while the group index increases. The decrease in n_{eff} can be understood from the fact that, at longer wavelengths, the optical mode becomes less tightly confined and spreads more into the surrounding cladding, whose refractive index is lower than that of silicon. Therefore, the modal effective index, which can be regarded as a weighted average experienced by the mode, decreases with wavelength.

The increase in group index is related to waveguide dispersion. The group index is given by

$$n_g = n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda}. \tag{16}$$

Since n_{eff} decreases with wavelength, $\frac{dn_{\text{eff}}}{d\lambda} < 0$, and thus the term

$$-\lambda \frac{dn_{\text{eff}}}{d\lambda} \tag{17}$$

is positive. In this waveguide, this positive dispersion term is large enough that n_g increases even though n_{eff} itself decreases.

The compact model of the waveguide can be expressed as

$$n_{\text{eff}}(\lambda) = 2.45 - 1.13(\lambda - 1.55) - 0.04(\lambda - 1.55)^2, \tag{18}$$

where λ is in μm.

B. Mach-Zehnder Interferometer

We then designed and simulated MZI1 with arm length differences of $\Delta L = 30$, 60, 90, and 120 μm using a Y-branch and a directional coupler. We also designed MZI2 with arm length differences of $\Delta L = 0$ and 60 μm using two Y-branches. In addition, two grating couplers were directly connected by a waveguide to evaluate the loss in the fabricated devices. The design layout is shown in Fig. 3. The transmission spectra of the MZIs are shown in Figs. 4 and 5. The simulated FSR values are shown in Table I. The group index can be extracted by

$$\text{FSR} = \frac{\lambda^2}{n_g \Delta L}. \tag{19}$$

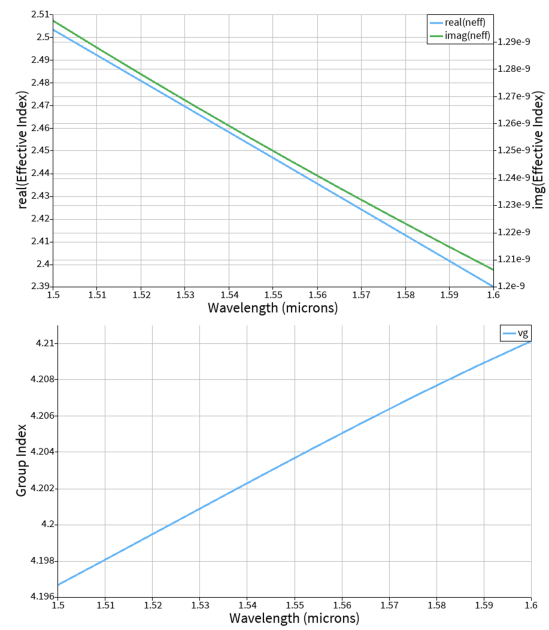

Fig. 2. Wavelength dependence of effective index and group index

TABLE I
FSR, AND GROUP INDEX OF MZI1.

ΔL (μm)	Simulated FSR (nm)	Measured FSR (nm)	n_g from measured FSR
30	19.05	19.55	4.096
60	9.53	9.686	4.134
90	6.35	6.438	4.146
120	4.76	4.808	4.164

From Fig. 5, it can be seen that the dominant loss contribution comes from the grating coupler insertion loss, since the MZI2-0 and GC-GC spectra nearly overlap over the whole wavelength range.

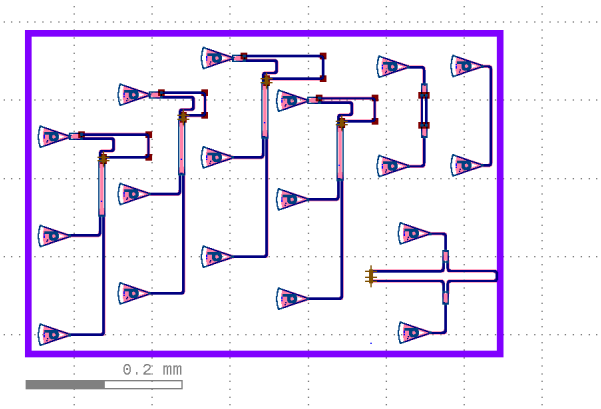

Fig. 3. Design layout.

IV. Measurement of MZI

A. Measurement results

The measured MZI spectra (Figs. 5 and 6) show clear interference fringes for all path-length differences, confirming proper device operation. The extracted FSR and calculated ng are shown in Table I. As expected, the fringe period decreases with increasing ΔL. The interference contrast is strongest in the central wavelength region, while it degrades toward the spectral edges, indicating a significant wavelength-dependent insertion-loss envelope dominated by the grating couplers and other passive components. This is also supported by the simulation and measurement results of the MZI-loss structure, which consists of two grating couplers directly connected by a waveguide, as well as the MZI2-0 and MZI2-60 devices fabricated using two Y-branch splitters. The relatively deep minima in the central region suggest that the arm loss imbalance is not severe. However, the two outputs are not perfectly complementary over the full wavelength range, which points to a non-ideal and wavelength-dependent splitter/combiner ratio, together with a small static phase bias introduced by fabrication imperfections.

The comparison among the GC-GC reference, the MZI-loss device (i.e., $\Delta L = 0$), and the MZI with $\Delta L = 60$ μm helps separate the baseline coupling loss from the intrinsic behavior of the interferometer. The GC-GC curve mainly represents the spectral response of the grating couplers and the measurement system, and therefore serves as a baseline envelope. The close agreement between the GC-GC and MZI-loss traces indicates that the additional excess loss introduced by the MZI structure itself is relatively small. In contrast, the MZI-60 μm trace exhibits clear periodic interference fringes while maintaining a similar upper envelope, suggesting that the added path-length difference mainly introduces interference behavior rather than significant extra propagation loss. The devices are measured with both Scylla and Iris testing stages. From data in Fig. 7, we can see that the data shows no significant differences while the loss spectra show slight difference in loss amplitude and wavelengh difference. This is likely due to the different coupling angles of fiber array.

B. Fabrication and measurement method

Fabricated is performed at one or more of these: Applied Nanotools and Washington Nanofabrication Facility. The following are the process descriptions.

Applied Nanotools, Inc. NanoSOI process:

The photonic devices were fabricated using the NanoSOI MPW fabrication process by Applied Nanotools Inc. (http://www.appliednt.com/nanosoi; Edmonton, Canada) which is based on direct-write 100 keV electron beam lithography technology. Silicon-on-insulator wafers of 200 mm diameter, 220 nm device thickness and 2 μm buffer oxide thickness are used as the base material for the fabrication. The wafer was pre-diced into square substrates with dimensions of 25×25 mm, and lines were scribed into the substrate backsides to facilitate easy separation into smaller chips once fabrication was complete. After an initial wafer clean using piranha solution (3:1 H_2SO_4:H_2O_2) for 15 minutes and water/IPA rinse, hydrogen silsesquioxane (HSQ) resist was spin-coated onto the substrate and heated to evaporate the solvent. The photonic devices were patterned using a JEOL JBX-8100FS electron beam instrument at The University of British Columbia. The exposure dosage of the design was corrected for proximity effects that result from the backscatter of electrons from exposure of nearby features. Shape writing order was optimized for efficient patterning and minimal beam drift. After the e-beam exposure and subsequent development with a tetramethylammonium sulfate (TMAH) solution, the devices were inspected optically for residues and/or defects. The chips were then mounted on a 4" handle wafer and underwent an anisotropic ICP-RIE etch process using chlorine after qualification of the etch rate. The resist was removed from the surface of the devices using a 10:1 buffer oxide wet etch, and the devices were inspected using a scanning electron microscope (SEM) to verify patterning and etch quality. A 2.2 μm oxide cladding was deposited using a plasma-enhanced chemical vapour deposition (PECVD) process based on tetraethyl orthosilicate (TEOS) at 300°C. Reflectrometry measurements were performed throughout the process to verify the device layer, buffer oxide and cladding thicknesses before delivery.

Washington Nanofabrication Facility (WNF) silicon photonics process

The devices were fabricated using 100 keV Electron Beam Lithography [1]. The fabrication used silicon-on-insulator wafer with 220 nm thick silicon on 3 μm thick silicon dioxide. The substrates were 25 mm squares diced from 150 mm wafers. After a solvent rinse and hot-plate dehydration bake, hydrogen silsesquioxane resist (HSQ, Dow-Corning XP-1541-006) was spin-coated at 4000 rpm, then hotplate baked at 80°C for 4 minutes. Electron beam lithography was performed using a JEOL JBX-6300FS system operated at 100 keV energy, 8 nA beam current, and 500 μm exposure field size. The machine grid used for shape placement was 1 nm, while the beam stepping grid, the spacing between dwell points during the shape writing, was 6 nm. An exposure dose of 2800 μC/cm^2 was used. The resist was developed by immersion in 25% tetramethylammonium hydroxide for 4 minutes, followed by

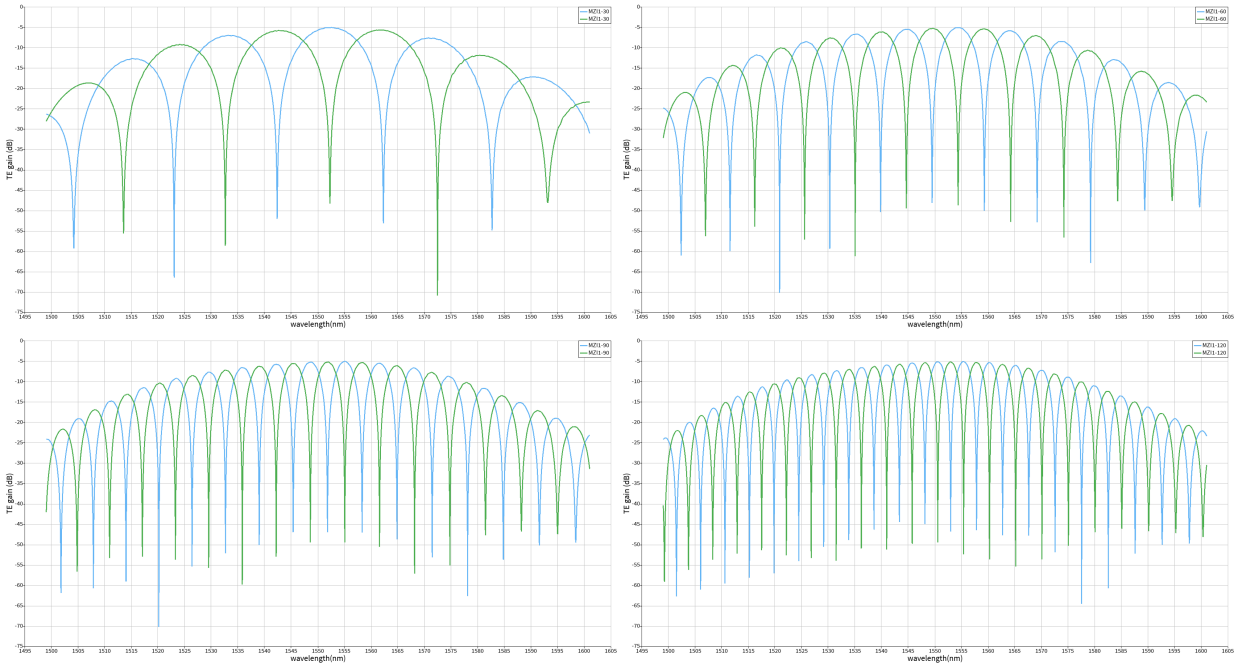

Fig. 4. Simulated transmission spectra of MZI1 with arm length differences of 30 μm (top left), 60 μm (top right), 90 μm (bottom left), and 120 μm (bottom right).

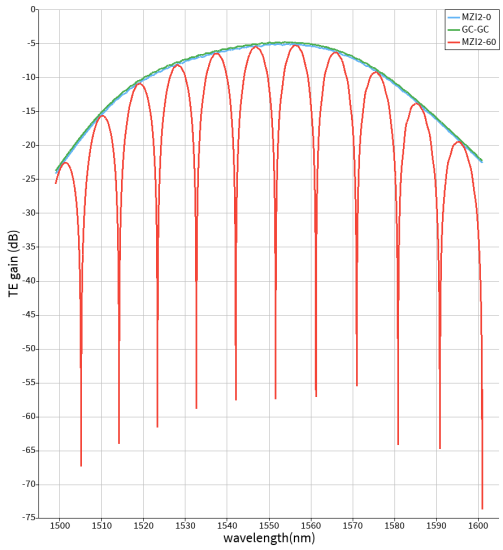

Fig. 5. Simulated transmission spectra of MZI2 with arm length differences of 0 μm, and 60 μm, and two grating couplers connected by a waveguide.

a flowing deionized water rinse for 60 s, an isopropanol rinse for 10 s, and then blown dry with nitrogen. The silicon was removed from unexposed areas using inductively coupled plasma etching in an Oxford Plasmalab System 100, with a chlorine gas flow of 20 sccm, pressure of 12 mT, ICP power of 800 W, bias power of 40 W, and a platen temperature of 20°C, resulting in a bias voltage of 185 V. During etching, chips were mounted on a 100 mm silicon carrier wafer using perfluoropolyether vacuum oil. Cladding oxide was deposited using plasma enhanced chemical vapor deposition (PECVD) in an Oxford Plasmalab System 100 with a silane (SiH_4) flow of 13.0 sccm, nitrous oxide (N_2O) flow of 1000.0 sccm, high-purity nitrogen (N_2) flow of 500.0 sccm, pressure at 1400 mT, high-frequency RF power of 120 W, and a platen temperature of 350°C. During deposition, chips rest directly on a silicon carrier wafer and are buffered by silicon pieces on all sides to aid uniformity.

Measurement description:

To characterize the devices, a custom-built automated test setup [2], [6] with automated control software written in Python was used [3]. An Agilent 81600B tunable laser was used as the input source and Agilent 81635A optical power sensors as the output detectors. The wavelength was swept from 1500 to 1600 nm in 10 pm steps. A polarization maintaining (PM) fibre was used to maintain the polarization state of the light, to couple the TE polarization into the grating couplers [4]. A 90° rotation was used to inject light into the TM grating couplers [4]. A polarization maintaining fibre array was used to couple light in/out of the chip [5].

V. Conclusion

In this article, we designed several Mach–Zehnder interferometers with different ΔL values using silicon strip waveguides. The waveguide and MZI simulations, as well as the measured MZI spectra, are presented. Both the measured and simulated spectra show that the dominant loss originates from the grating couplers. The small difference between the simulated and measured free spectral range (FSR) indicates that the fabrication variation is small.

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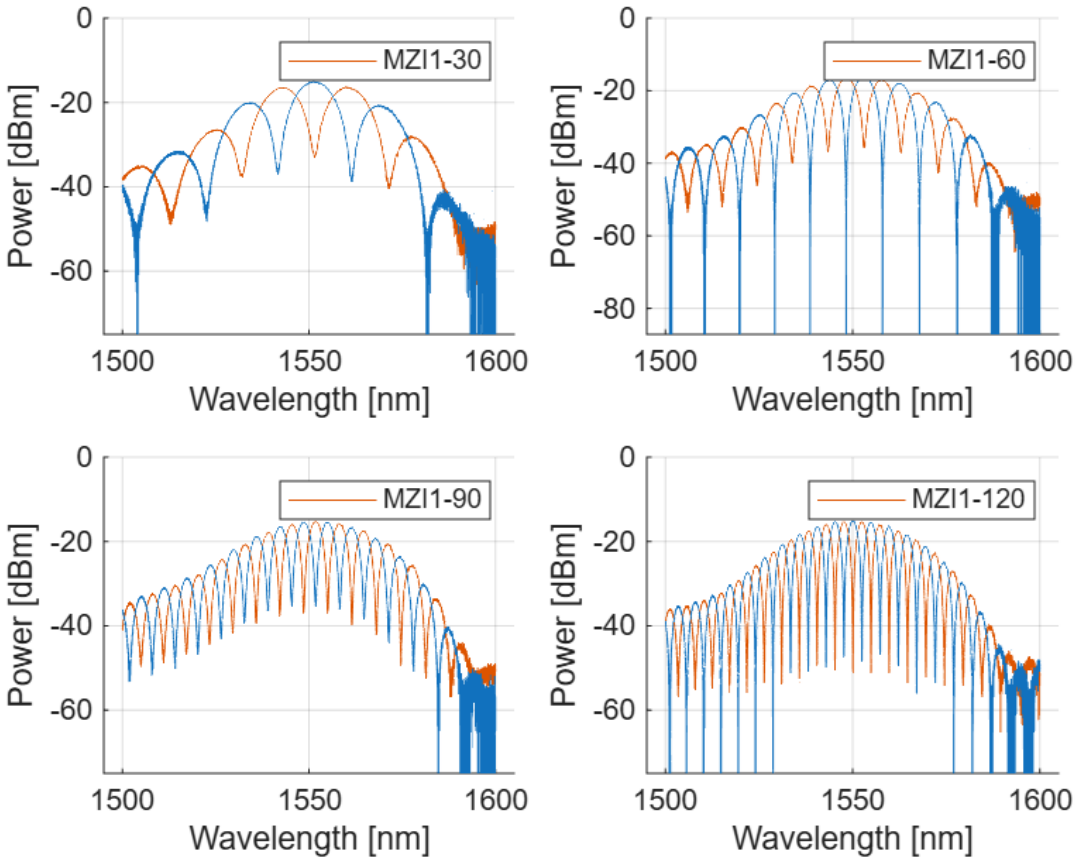

Fig. 6. Measured transmission spectra of MZI1 with arm length differences of 30 μm (top left), 60 μm (top right), 90 μm (bottom left), and 120 μm (bottom right).

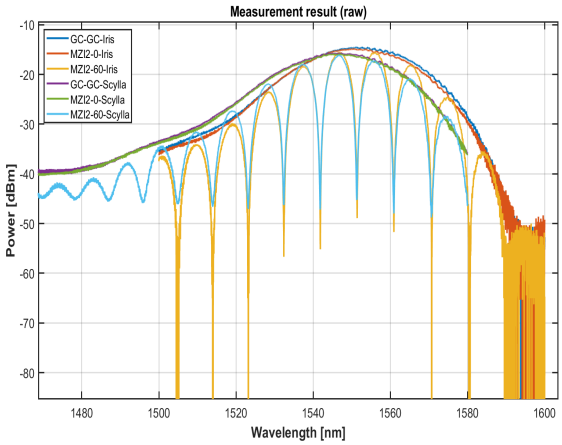

Fig. 7. Measured transmission spectra of MZI2 with arm length differences of 0 μm, and 60 μm, and two grating couplers connected by a waveguide. The devices are measured with two testing stages Iris and Scylla

supported by the Natural Sciences and Engineering Research Council of Canada (NSERC) Silicon Electronic-Photonic Integrated Circuits (SiEPIC) Program. The devices were fabricated by Richard Bojko at the University of Washington Washington Nanofabrication Facility, part of the National Science Foundation's National Nanotechnology Infrastructure Network (NNIN), and Cameron Horvath at Applied Nanotools, Inc. Mateo Branion-Calles performed the measurements at The University of British Columbia. We acknowledge Lumerical Solutions, Inc., Mathworks, Mentor Graphics, Python, and KLayout for the design software.

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